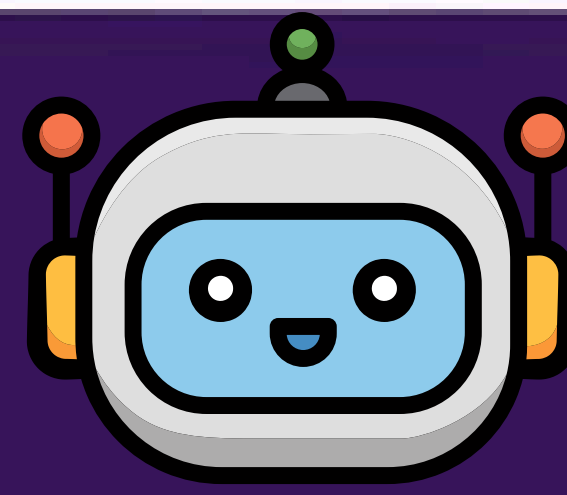


TEAM MEMBER
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STAB-E



(Self-balancing Two-wheel ARM Bot Entity)

Cascaded PID control with manipulator Arm for Agirculture Automation
Designed for stability, precision control, and autonomous mobile manipulation



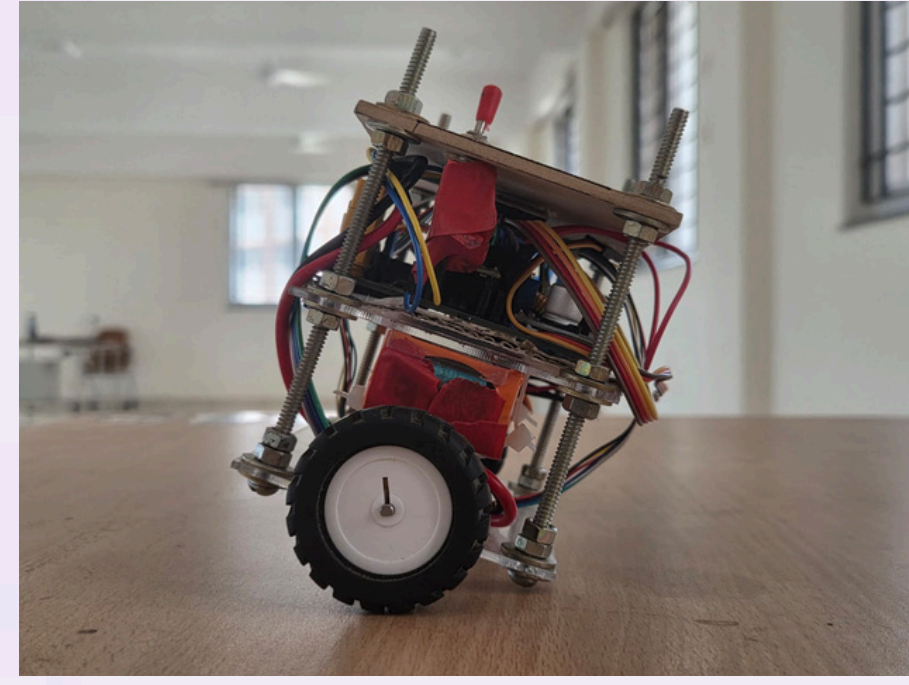
PROJECT OVERVIEW

- Self balancing robot based inverted pendulum principle
- Built for eYRC robotics competition , hosted by IIT Bombay.
- Capabilities :**
 - balancing in one place
 - Manipulator arm for payload tasks
 - wireless connection
 - 3.5 - 4 hrs of battery life

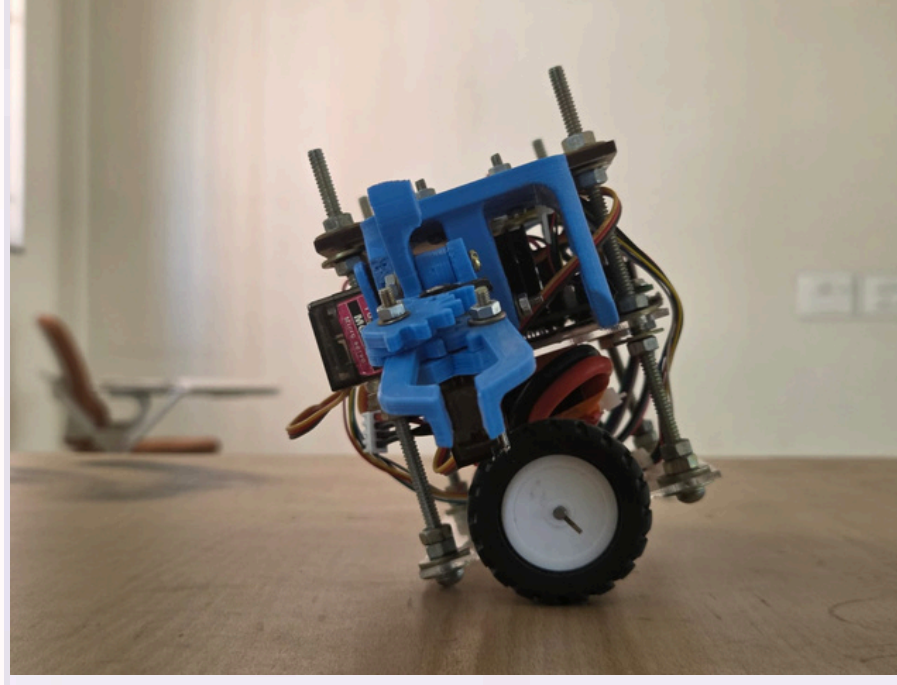
Team ID : KB#2155
Semi-Finalist
(Stage 2)

ROBOT AT A GLANCE

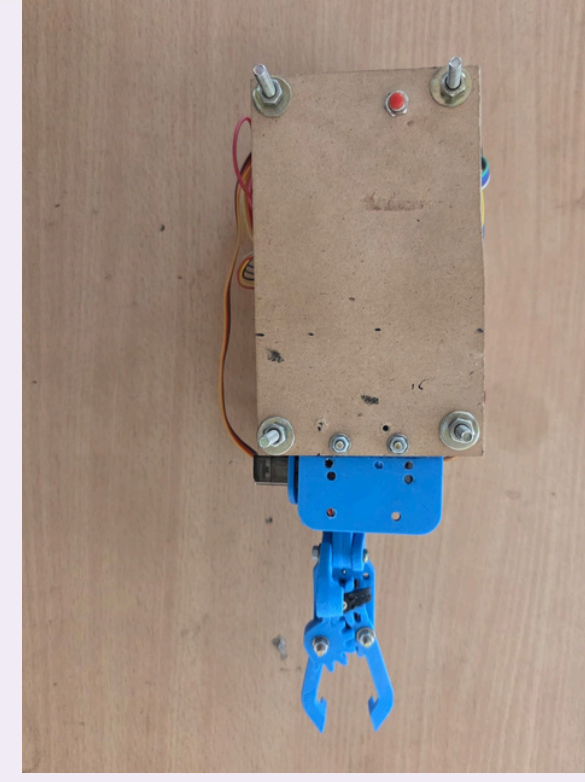
LEFT VIEW



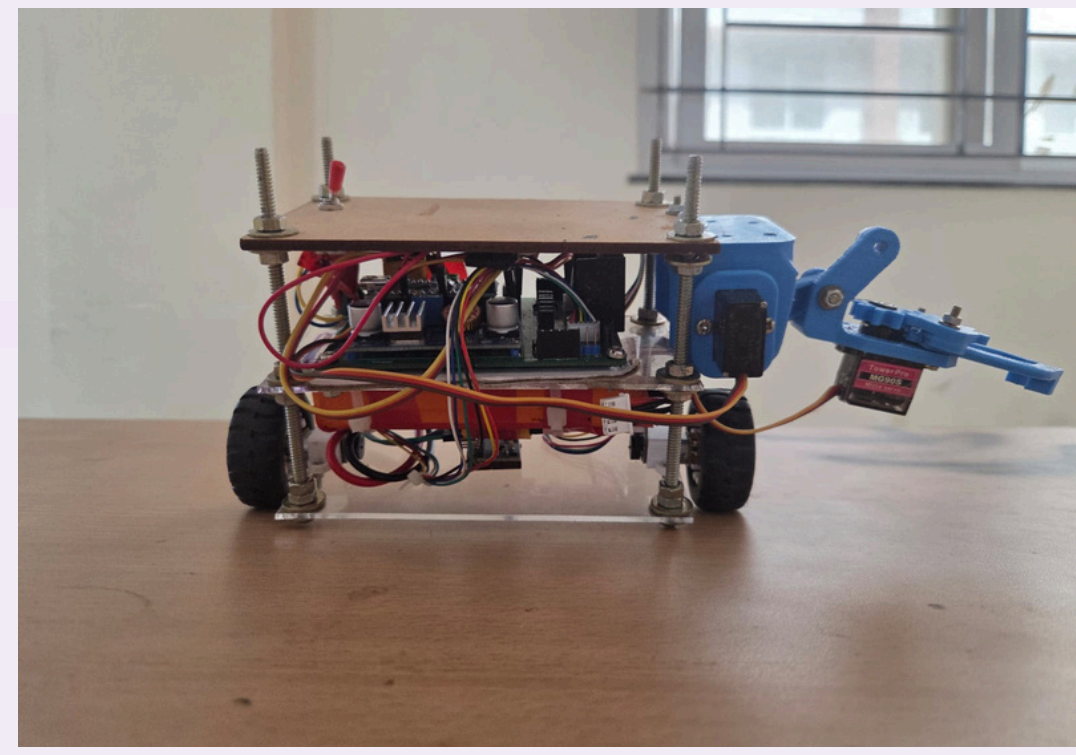
RIGHT VIEW



TOP VIEW



FRONT VIEW



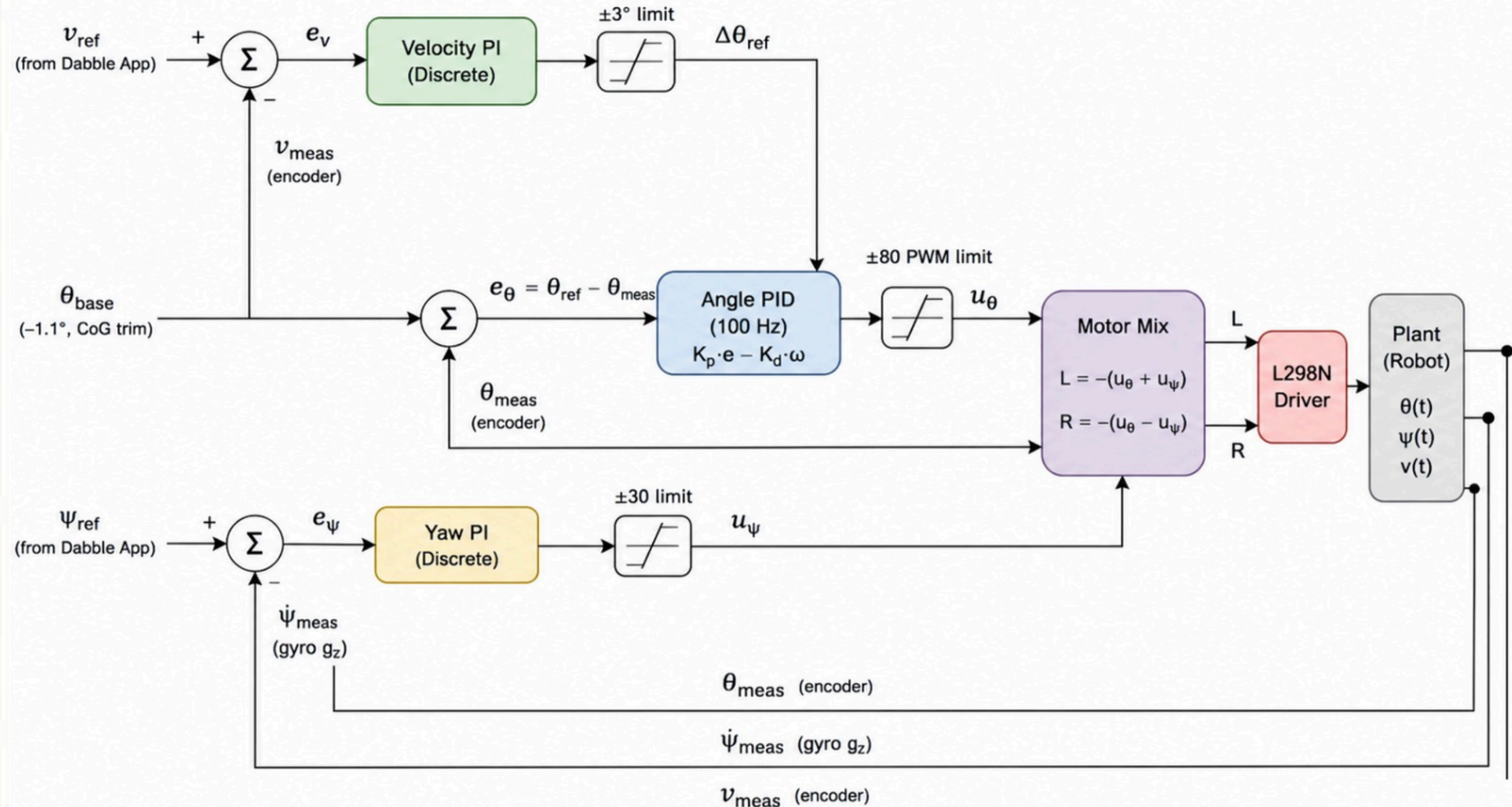
BACK VIEW



HARDWARE SPECIFICATION

Chassis	3-layer MDF (laser-cut)
Wheel diameter	43 mm
Wheel base	~100 mm
Drive motors	2× N20 300 RPM DC gear motors
Encoder resolution	~6170 ticks / revolution
Arm	Blue PLA 3D-printed, 2× TowerPro MG90S servos
Total weight (est.)	~350 g
Microcontroller	Arduino Nano (ATmega328P)
IMU	MPU6050 (GY-521 module)
Motor driver	L298N dual H-bridge
Power	7.4 V 2S LiPo, ~1200 mAh
Bluetooth	HC-05 module
Servo	TowerPro MG90S × 2

CONTROL ARCHITECTURE



- Key Decisions :**
 - D-term uses raw gyro rate (not differentiated angle)
 - Outer loop outputs an angle offset, not direct PWM
 - Yaw uses gyro gz, not encoders
 - I2C at 400 kHz (Fast Mode)

SENSOR FUSION

Complementary Filter
(9 - Pitch Imgin)

$$\theta[n] = 0.96 \times (\theta[n-1] + \omega[n] \times dt) + 0.04 \times \theta_acc$$

Why Complementary Filter?

Gyro - Short-term accuracy.
Accelerometer - long-term stability.
Cutoff Frequency = 0.637Hz.
Efficient, Reliable and drift resistant.

IMU Details

- MPU6050 (16-bit ADC)
- Gyro Range: $\pm 2000^\circ/\text{s}$
- Accel Range: $\pm 8g$
- 12C Speed: 400KHz
- Sampling Rate: 100HZ

PID Tuning Results

Final PID Gains

Controller	Kp	Ki	Kd	Rate
Angle(Inner)	45.0	0.0	0.8	100Hz
Velocity (Outer)	0.025	0.01	-	20Hz
Yaw(PI)	1.5	0.3	-	20Hz

Highlights

- System remains stable even with arm inertia.
- Kd increased from 0.5 to 0.8 to handle extra load.
- Zero steady-state error in tilt and yaw.

Tuning using Ziegler-Nichols, then refined manually for real-world stability

ELECTRICAL CONNECTIONS

Pin Map (Arduino Nano)

Motor Driver (L298N)

D6(PWM) ENA- Left Motor Speed
A2 IN1 - Left Motor Dir
A3 IN2 - Left Motor Dir
D5(PWM) ENB- Right Motor Speed
D9 IN3 - Right Motor Dir
D4 IN4 - Right Motor Dir

Encoders

D2 Left Encoder A (INT0 -Reliable)
D3 Left Encoder B (INT1 -Reliable)
D7 Left Encoder A (PCINT -Unreliable)
D8 Left Encoder B (PCINT -Unreliable)

Peripherals

A4 (SDA) MPU6050 SDA
A5 (SCL) MPU6050 SCL
A1 BUZZER
A0 Servo(Arm)
D12(Rx) HC-05 TX ->nano RX
D13(Tx) HC-05 TX ->nano TX

Note :

- HC-05 moved to pins 12/13 to avoid PCINT conflict with encoders.
- Right motor polarity inverted on chassis -> handled in firmware.
- baseTargetAngle = -1.1° compensates for CoG offset.

⚠️Right encoder PCINT issue: At N20 motor speed (~300 RPM × gear ratio), PCINT interrupts miss ~96% of pulses due to timing conflicts with the PWM timer. Yaw is controlled via gyro_gz instead.

Documentation and Source code

Scan QR to access Repository Contents



Full source code, Arduino firmware, CoppeliaSim simulation & circuit documentation. Cascaded PID | Startup guide Demo Vedio (Hardware and Simulation)
Team KB#2155 — eYRC 2025-26, IIT Bombay.

TASK SUMMARY

Hardware Implementation | Simulation

Task 1 —
Simulation: Basic Balancing (CoppeliaSim)
Implemented complementary filter and inner angle PID loop in Pythob.. Robot balances in place with no external disturbances.

Task 2 —
Simulation: Motion Control (Webots)
Added outer velocity loop and yaw control. Robot responds to keyboard commands — forward, backward, and turn in simulation.

Task 3 —
Hardware: Full Balancing Demo, All three control loops running on Arduino Nano hardware. Robot balances, carries payload using the servo arm, and recovers from pushes.

APPLICATIONS & PURPOSE

Application	Purpose
- Agricultural Automation Robots - Terrain Balancing Platforms - Mobile manipulation Systems	- Design a stable two-wheeled robot using cascaded PID - Integrate balancing with manipulation - Develop an efficient embedded control system - Ensure robust real-world performance

CALIBRATIONS -Key Procedures

Encoder Calibration	Motor Calibration
- Rotate output shaft exactly one full turn. - Count / 4 = Encoder CPR. - CPR × Gear Ratio = TICKS_PER_REV. - Current Value: ~6170 ticks/rev	- Run each motor at PWM = 100. - Measure RPM / ticks per second. - scale = ref_velocity / measured_velocity. - Update leftMotorScale / rightMotorScale.

STARTUP -Key Procedures

Gyro Calibration

- Place robot on flat surface (still).
- Wait for single beep -> calibration starts.
- Do NOT move robot (~4 seconds).
- Offsets stored: gyro_offset & gyro_offset_z.